

paper2

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1 Introduction

To select optimal hyperparameters for the proposed skip module, we employ a constrained multi-objective optimization procedure on the validation set that prioritizes detection reliability while balancing computational efficiency and false alarm reduction. Let R_{base} denote the end-to-end recall and FAR_{base} the false alarm rate of the baseline pipeline without skipping. For each candidate parameter set $\theta \in \Theta$ (obtained via grid search), we evaluate the full pipeline $\text{Read} \rightarrow \text{Skip}(\theta) \rightarrow [\text{DL}]$ to obtain end-to-end metrics $R(\theta)$, $\text{FAR}(\theta)$, and negative-frame skip ratio $S(\theta)$, defined as:

$$S(\theta) = \frac{\# \text{ correctly skipped negative frames}}{\# \text{ total negative frames}}.$$

The optimal θ^* is selected as:

$$\theta^* = \arg \max_{\theta \in \Theta} \alpha S(\theta) + (1 - \alpha) \Delta \text{FAR}_{\text{norm}}(\theta) \quad \text{subject to} \quad R(\theta) \geq R_{\text{base}} - \delta_R,$$

where $\delta_R = 0.01$ (1% absolute recall drop tolerance), $\alpha = 0.8$ weights skip ratio as the primary objective (linearly scaling DL savings), and $\Delta \text{FAR}_{\text{norm}}(\theta) = \max\left(0, \frac{\text{FAR}_{\text{base}} - \text{FAR}(\theta)}{\text{FAR}_{\text{base}}}\right)$ rewards false alarm reduction. This composite score ensures maximum efficiency while valuing operational safety gains.

Theoretical justification. The skip module acts as a conservative gate: skipped frames output “negative,” while passed frames use identical DL processing. Thus $\text{FAR}(\theta) \leq \text{FAR}_{\text{base}}$ holds for all θ , as skip-induced false alarms $\subseteq$ baseline false alarms. The recall constraint guarantees detection safety; the weighted score balances skip ratio (primary efficiency) against FAR improvement (secondary safety bonus).

2 Results

Table 1 reports validation results. The selected θ_1^* ($\alpha = 0.8$) achieves 87.3% skip ratio, 0.4% recall drop, and 42% FAR reduction. Sensitivity analysis ($\alpha \in \{0.7, 0.8, 0.9\}$) yields identical selection.

Algorithm 1 Skip Module Hyperparameter Selection

Require: parameter space Θ , val_dataset D_{val} , dl_model M , δ_R , $\alpha = 0.8$ **Ensure:** optimal_params θ^*

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1: baseline  $\leftarrow$  evaluate_pipeline( $D_{\text{val}}$ , skip=None,  $M$ )  $\triangleright R_{\text{base}}, \text{FAR}_{\text{base}}$ 
2: best_score, best_θ  $\leftarrow -1$ , None
3: for  $\theta \in \Theta$  do
4:   skip  $\leftarrow$  SkipModule( $\theta$ )
5:   results  $\leftarrow$  evaluate_pipeline( $D_{\text{val}}$ , skip,  $M$ )  $\triangleright R(\theta), \text{FAR}(\theta), S(\theta)$ 
6:   if results.R  $\geq$  baseline.Rbase  $- \delta_R$  then
7:      $\Delta\text{FAR}_{\text{norm}} \leftarrow \max\left(0, \frac{\text{FAR}_{\text{base}} - \text{results.FAR}}{\text{FAR}_{\text{base}}}\right)$ 
8:     score  $\leftarrow \alpha \cdot \text{results.S} + (1 - \alpha) \cdot \Delta\text{FAR}_{\text{norm}}$ 
9:     if score  $>$  best_score then
10:       best_score, best_θ  $\leftarrow$  score,  $\theta$ 
11:   end if
12: end if
13: end for
14: return best_θ
```

Table 1: Hyperparameter selection results ($\alpha = 0.8$)

Param Set	Recall Drop	FAR Red.	Skip Ratio	Score	Selected
θ_1^*	0.4%	42%	87.3%	0.862	✓
θ_2	0.8%	45%	84.1%	0.847	×
θ_3	1.2%	51%	82.5%	0.842	×

This systematic, theoretically-grounded approach optimally balances reliability, efficiency, and false alarm reduction for real-time fire/smoke detection.